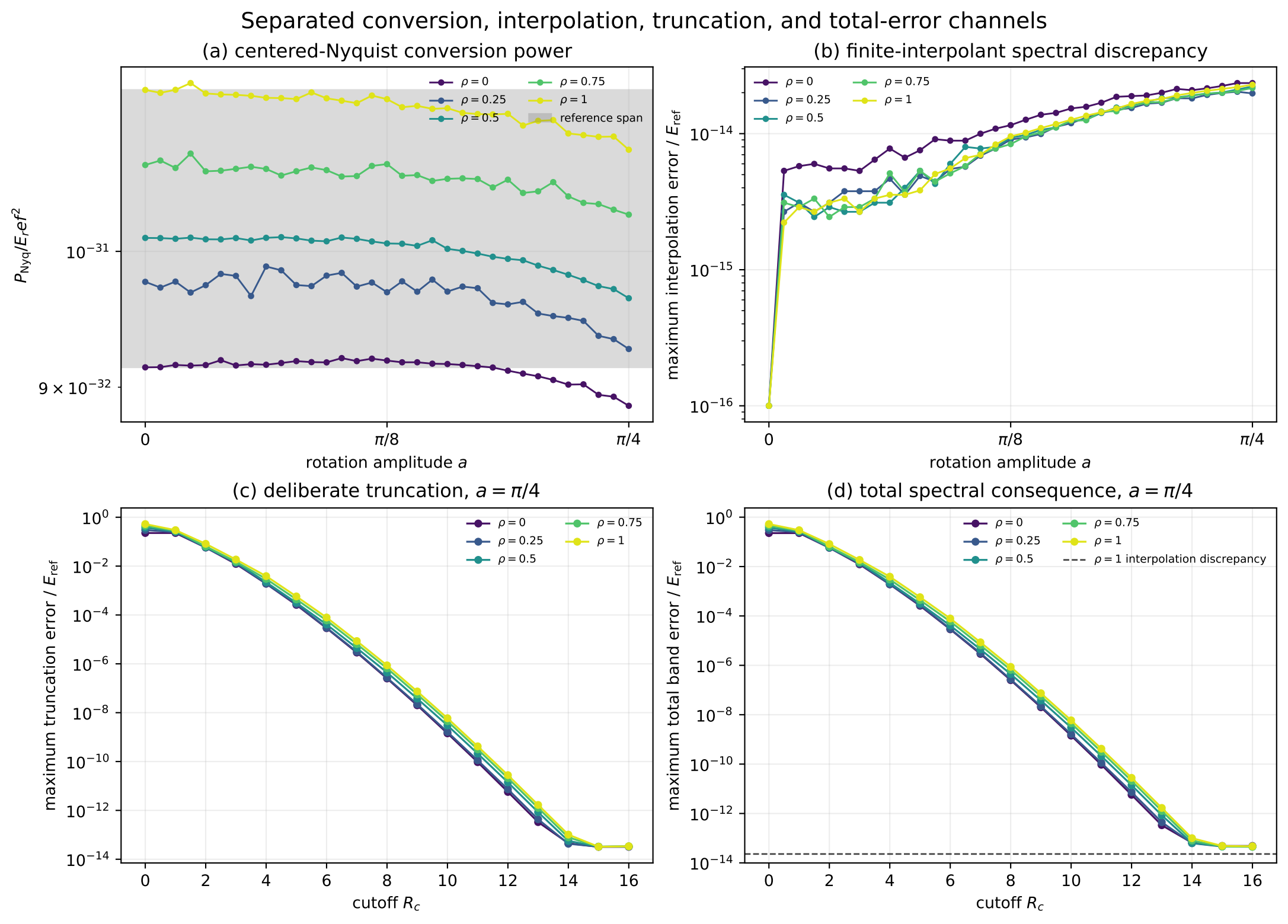
Separated conversion, interpolation, truncation, and total-error channels
(a) centered-Nyquist conversion power
$P_{Nyq}/E_r ef^2$
10^{-31}
9×10^{-32}
$\rho = 0$
$\rho = 0.25$
$\rho = 0.5$
$\rho = 0.75$
$\rho = 1$
reference span
0
π/8
π/4
rotation amplitude a
(b) finite-interpolant spectral discrepancy
maximum interpolation error / E_{ref}
10^{-14}
10^{-15}
10^{-16}
$\rho = 0$
$\rho = 0.25$
$\rho = 0.5$
$\rho = 0.75$
$\rho = 1$
0
π/8
π/4
rotation amplitude a
(c) deliberate truncation, $a = \pi/4$
maximum truncation error / E_{ref}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
$\rho = 0$
$\rho = 0.25$
$\rho = 0.5$
$\rho = 0.75$
$\rho = 1$
0
2
4
6
8
10
12
14
16
cutoff R_c
(d) total spectral consequence, $a = \pi/4$
maximum total band error / E_{ref}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
$\rho = 0$
$\rho = 0.25$
$\rho = 0.5$
$\rho = 0.75$
$\rho = 1$
$\rho = 1$ interpolation discrepancy
0
2
4
6
8
10
12
14
16
cutoff R_c